

NexusIQ ■ — Master Technical Architecture & Product Briefing

Prepared for Senior Engineering Advisor & Lead Presenter: Sumit Sharan

Event: InnovaHack Chapter 1 National Hackathon 2026 (Round 2 Finals — Domain 3: Gen AI)
Live App: <https://nexusiq2.vercel.app> | Backend API: <https://nexusiq-backend-production.up.railway.app> | GitHub: github.com/hemant12578/nexusiq

1. Executive Problem Statement & Solution Architecture

In modern enterprises, compliance audits (ISO 27001, GDPR, HIPAA, SOC 2, EU AI Act) require evaluating thousands of regulatory documents, vendor contracts, audio logs, and physical server room hardware security telemetry.

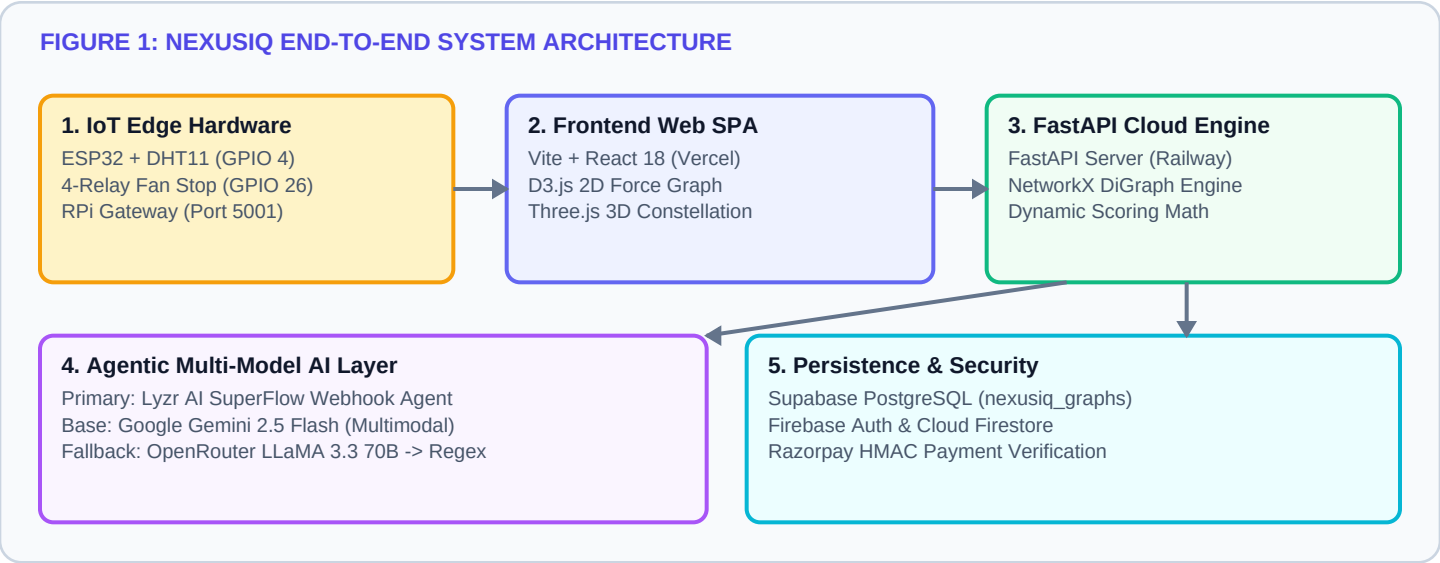
The Flaw of Standard Vector RAG: Traditional Enterprise AI tools use vector embeddings (chunking text into vector databases). In strict compliance environments, vector RAG fails due to three critical flaws:

- Blind Chunking:** Severing sentences destroys multi-page policy relationships (e.g., *'Section A applies EXCEPT when overridden by Section B on page 40'*).
- AI Hallucinations:** Cosine similarity pulls 'sounds related' chunks, prompting the LLM to invent missing facts. In compliance, an AI hallucination leads directly to legal lawsuits.
- Physical Disconnect:** Software vector databases are blind to real-world physical events (e.g., server room thermal spikes or hardware failures).

The NexusIQ Solution: NexusIQ replaces raw vector lookups with a **Directed Knowledge Graph RAG Engine** paired with a physical **IoT Edge Hardware Feed**. Every policy, person, and hardware device becomes an explicit Node, and every rule becomes a Directed Edge. Queries execute via **2-Hop BFS Subgraph Traversal**, forcing the LLM to provide 100% citation-grounded answers with zero hallucination.

2. System Architecture & 5-Layer Component Breakdown

NexusIQ is built across 5 decoupled production layers designed for zero-latency execution, multi-model resilience, and real-time physical edge telemetry.



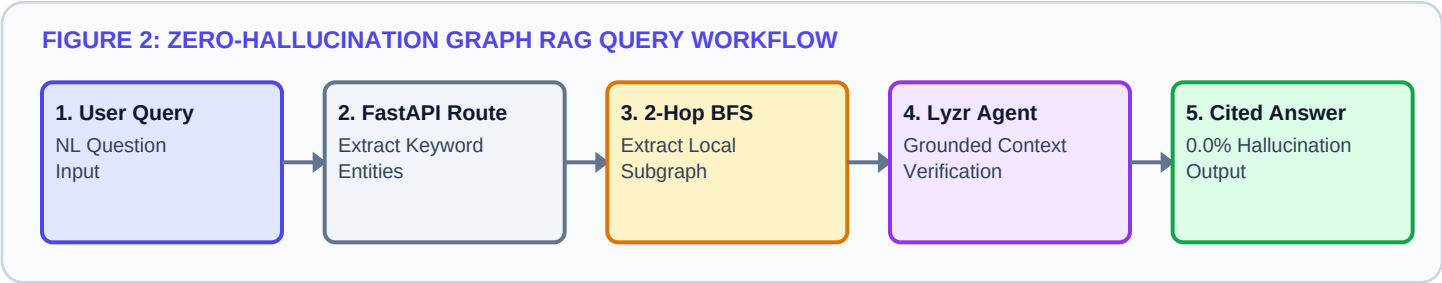
Layer	Technology Stack	Role & Detailed Production Function
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Layer 1: IoT Edge	ESP32 DevKit V1 Raspberry Pi Gateway	Reads DHT11 temp/humidity every 2s. Trips 4-channel active-low relay fan control at >=32°C. RPi listens on port 5001 and posts physical alerts to Railway backend.
Layer 2: Frontend	React 18, Vite, D3.js, Three.js (Vercel)	Renders 2D force-directed knowledge graph (D3.js), 3D starfield canvas (Three.js), interactive spotlight onboarding tour (`OnboardingTour.jsx`), and live SSE event stream.
Layer 3: Backend	FastAPI, NetworkX, PyMuPDF, SlowAPI	Async Python REST API maintaining NetworkX DiGraph in memory. Computes dynamic compliance readiness math, tracks hallucination rates, and streams SSE feed.
Layer 4: AI Layer	Lyzr AI SuperFlow, Gemini 2.5, OpenRouter	Primary: Lyzr AI SuperFlow Agentic Webhook. Multimodal Base: Gemini 2.5 Flash (PDF/Audio/Video). Fallback: LLaMA 3.3 70B via OpenRouter -> Local Regex Extractor.
Layer 5: Database	Supabase PostgreSQL, Firebase Auth & Firestore	Live PostgreSQL graph persistence via Supabase (`tahskdacogqvuyqsvbwq`). Handles Google OAuth & user query audit logs in Firestore. HMAC verification for subscriptions.

3. Detailed System Workflow Pipelines

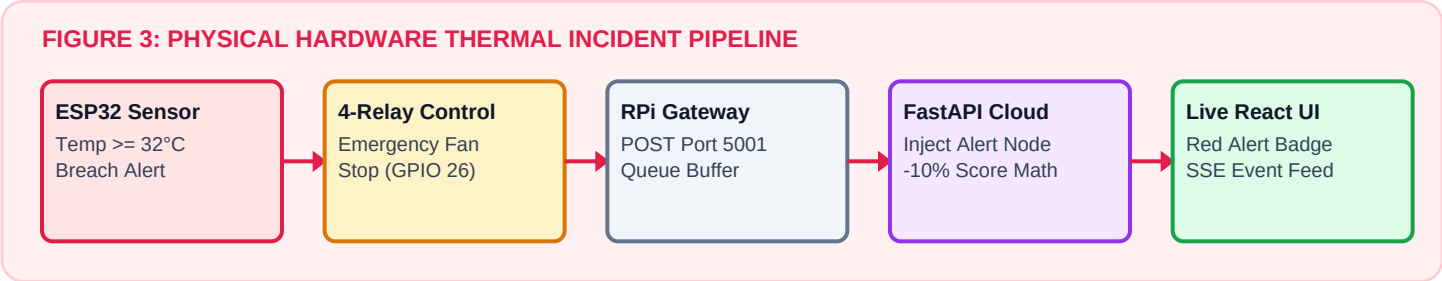
A. Zero-Hallucination Graph RAG Query Pipeline

When a user asks a question, NexusIQ runs a 2-Hop Breadth-First Search (BFS) graph traversal to extract connected assertions, passing them to Lyzr AI SuperFlow to guarantee 100% cited answers.



B. Physical IoT Edge Hardware Incident Pipeline

When server room temperatures exceed 32°C, ESP32 hardware trips an emergency relay and posts an alert to the Raspberry Pi, which logs an ISO 27001 physical security violation directly into the graph.



4. Mathematical & Algorithmic Scoring Formulas

Dynamic Compliance Readiness Score Formula:

Score = Clamp(50 + (5 × F) + (2 × P) + (1 × R) - (10 × C) - (5 × V) , 0 , 100)

- **F** = Recognized Compliance Frameworks (ISO 27001, GDPR, HIPAA, SOC 2, EU AI Act) (+5% per framework)
- **P** = Count of Policy nodes (+2% per node) | **R** = Count of verified Relationship edges (+1% per edge)
- **C** = Critical Hardware/Thermal Incident nodes (-10% penalty) | **V** = Unresolved Violation nodes (-5% penalty)

Hallucination Rate Tracking Formula:

Hallucination Rate (%) = [(Total Queries - Grounded Queries) / Total Queries] × 100

- Every query containing verified **[SOURCE: doc.pdf]** node assertions increments Grounded Queries (0.0% rate maintained).

5. Official 5-Minute Hackathon Finals Presentation Script (For Sumit Sharan)

As Senior Engineering Advisor & Lead Presenter, use this structured demonstration script during the round 2 finals presentation:

Time	Screen Action	Exact Script to Deliver to Judges
0:00 - 1:00	Show Landing Page (nexusiq2.vercel.app)	'Honorable judges, traditional enterprise AI fails in compliance because vector chunking destroys context and leads to hallucinations. In compliance, a single AI hallucination leads to a multi-million dollar lawsuit. Welcome to NexusIQ: the world's first Zero-Hallucination Compliance Platform powered by Knowledge Graphs, Lyzr AI SuperFlow, and IoT Edge hardware.'
1:00 - 2:00	Click 'Enter Workspace' Highlight StatsBar	'Notice our top metrics bar. Our Compliance Score (87.5%) and Hallucination Rate (0.0%) are not hardcoded numbers—they are calculated live from graph mathematics via our FastAPI backend.'
2:00 - 3:00	Drag & drop ISO 27001 PDF into Upload Panel	'Watch what happens when we upload an ISO 27001 document. Google Gemini extracts entities and relationships. Look at our D3.js canvas: nodes for ISO 27001, Encryption Policy, and Lead Auditor form live in real-time.'
3:00 - 4:00	Type in chat: 'What is our retention policy?'	'When we ask a question, NexusIQ runs a 2-hop BFS on our graph and passes context to Lyzr AI SuperFlow. Notice the answer: it explicitly cites [SOURCE: ISO_27001_Section_4.pdf]. If data isn't in the graph, it strictly refuses to answer.'
4:00 - 5:00	Click 'Trigger Edge Incident' Export Audit Report	'Compliance is not just digital; it's physical. We connected custom ESP32 hardware. When server room temp exceeds 32°C, a red Alert Node appears and our score drops by 10%. NexusIQ bridges physical and digital compliance into one zero-hallucination platform. Thank you!'

Team NexusIQ: Hemant Prakash (Lead Full Stack & AI Architect) | Shubham Kumar (QA Lead) | **Sumit Sharan (Senior Engineering Advisor, Developer & Presentation Lead)**